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Class:SY B-AIDS

ROLL NO:-B-40

SUB:-PDS

Experiment 10

Aim

To study and understand Version Control using Git and GitHub, and perform basic operations like adding, committing, and pushing files to a repository.

Theory

Version Control System (VCS) is used to track changes in files and manage different versions of a project. It helps multiple developers work together efficiently.

Git is a distributed version control system that tracks changes locally on your system.

GitHub is a cloud-based platform that hosts Git repositories and allows collaboration.

Basic workflow in Git:

- Create repository
- Add files
- Stage files (git add)
- Commit changes (git commit)
- Push to remote repository (git push)

Screenshots:



shifanshaikh-9030

Edit profile

🔗 Joined 2 days ago

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PYTHON-FOR-DATA-SCIENCE- Public

4 contributions in the last year

Contribution settings



Contribution activity

Year: 2026

May 2026

📁 Created 2 commits in 1 repository
shifanshaikh-9030/PYTHON-FOR-DATA-SCI...
2 commits

📁 Created their first repository

May 5



🔗 Joined GitHub



Seeing something unexpected? Take a look at the [GitHub profile guide](#).



... / P...



Code

Issues

Pull requests

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Apr 30 - May 7 ▾

Commits on May 5, 2026

Add files via upload

Verified

...



shifanshaikh-9030 authored 2 days ago

Initial commit

Verified

...



shifanshaikh-9030 authored 2 days ago

Seeing something unexpected? Take a look at the [GitHub commits guide](#).

github.com/shifanshaikh-9030/



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main Go to file + <> Code

shifanshaikh-9030	Add files via upload	a0ba7da · 2 days ago
EXP1_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP2_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP3_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP4_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP5_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP6_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP7_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP8_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
EXP9_B3_B-40_SHIFAN SHAIKH...	Add files via upload	2 days ago
README.md	Initial commit	2 days ago

README

PYTHON-FOR-DATA-SCIENCE-

About

No description, website, or topics provided.

Readme

Activity

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Releases

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Contributors 1

shifanshaikh-9030



739 KB

03/05/2026, 02:20

EXP8 - Colab

NAME: SHIFAN NASIR SHAIKH

ROLL NO: B-40

BRANCH: AI&DS(B)

BATCH: B3

AIM

Implement and analyze various methods of studying correlation and regression including Karl Pearson's coefficient and simple linear regression.

Theory

Correlation measures the strength and direction of the relationship between two variables.

Regression is used to predict the value of one variable based on another.

Karl Pearson's coefficient (r) is the most common measure of correlation and ranges from -1 to $+1$:

$+1 \rightarrow$ Perfect positive correlation $-1 \rightarrow$ Perfect negative correlation $0 \rightarrow$ No correlation

Simple Linear Regression models the relationship between two variables using a straight-line equation:

$$y = mx + c$$

where:

m = slope c = intercept

1. What is correlation? How is it different from regression?
Correlation shows the relationship between variables, while regression is used for prediction.

Correlation $\rightarrow$ symmetric (X and Y are equal) Regression $\rightarrow$ asymmetric (predict Y from X)

Code (Correlation Calculation Example)

```
import numpy as np

# Sample data
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 6, 8, 10])
```

https://colab.research.google.com/drive/11nknEmg7l0Z9mZe_KA5GqLSpxuJO7g1#scrollTo=qxv1c365dipi&printMode=true

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EXP8 - Colab

```
# Correlation
corr = np.corrcoef(x, y)[0,1]

print("Correlation coefficient:", corr)

Correlation coefficient: 0.9999999999999999
```

Questions and Answers:

1. What is the difference between Git and GitHub?

2. Feature	Git	GitHub
Definition	Git is a version control system	GitHub is an online platform for hosting Git repositories
Purpose	Tracks changes in code/files	Stores projects online and helps teams collaborate
Works On	Your local computer	Cloud / Internet
Created By	Linus Torvalds	GitHub
Internet Needed	No, can work offline	Yes, mainly online
Main Use	Managing versions of files	Sharing and collaborating on projects
Commands/Features	git add, git commit, git push	Pull requests, issues, repositories, collaboration tools
Example	Saving versions of a project	Uploading project for others to access

2. Explain the difference between git add and git commit

`git add` and `git commit` are two important Git commands used for version control. The `git add` command is used to move changed files to the staging area, which means Git prepares those files to be saved. It does not permanently save the changes; it only selects which changes should be included in the next version. On the other hand, the `git commit` command is used to permanently save the staged changes into the Git repository with a message describing the changes made. In simple words, `git add` prepares the changes, while `git commit` stores them as a new version in the project history.

3. What is the staging area in Git?

The staging area in Git is an intermediate space where changes are prepared before they are permanently saved in the repository. When a file is modified, Git does not directly save the changes. First, the `git add` command is used to place the selected changes into the staging area. After reviewing the changes, the `git commit` command is used to save them permanently. The staging area helps developers

choose which changes should be included in the next commit, making version control more organized and flexible.

4. What happens if you modify a file after committing it?

If you modify a file after committing it, Git recognizes the file as “modified” because the changes are different from the last saved version in the repository. The previous commit remains unchanged, and the new modifications are not automatically saved. To include the new changes in the project history, you must again use `git add` to stage the file and then use `git commit` to create a new commit. This allows Git to keep track of every version of the file separately.

5. Difference between git push and git pull

`git push` and `git pull` are Git commands used to synchronize changes between a local repository and a remote repository such as [GitHub](#). The `git push` command is used to upload local commits from your computer to the remote repository so that others can access the updated project. In contrast, the `git pull` command is used to download the latest changes from the remote repository to your local system. In simple terms, `git push` sends changes to the server, while `git pull` brings changes from the server to your computer.

6. What is Version Control? Differentiate between Centralized and Distributed VCS

Version Control is a system that helps developers track and manage changes made to files, documents, or source code over time. It allows multiple users to work on the same project, keeps a history of modifications, and helps restore previous versions if needed. Version control systems improve collaboration, reduce conflicts, and make software development more organized and efficient.

The two main types of Version Control Systems are Centralized Version Control System (CVCS) and Distributed Version Control System (DVCS). In a Centralized VCS, there is a single central server that stores all project files and version history, and users must connect to this server to access or update files. If the central server fails, the entire project may become inaccessible. Examples include [Apache Subversion \(SVN\)](#) and [CVS](#). On the other hand, in a Distributed VCS, every user has a complete copy of the repository along with its history on their local system. This allows users to work offline and provides better backup and reliability. [Git](#) is the most popular example of a Distributed VCS.